

# Questions for Image Processing (CSC321)(Chapter-wise)

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## Unit 1: Introduction

### 5-Mark Questions

1. Define a digital image. Explain the representation of a digital image in spatial and matrix form.
2. Explain the fundamental steps of digital image processing with a labeled block diagram.
3. Describe brightness adaptation and discrimination in human visual perception.
4. What is sampling and quantization? Explain spatial and gray-level resolution.
5. Define adjacency and connectivity. Explain different distance measures between pixels.

### 10-Mark Questions

1. Explain the human visual system and image formation in the eye with a neat diagram.
2. Discuss sampling and quantization in detail. Also explain how resolution affects image quality.
3. Describe pixel neighbors, regions, boundaries, and connectivity with suitable examples.
4. Explain the complete fundamental steps in digital image processing with applications in detail.

## Unit 2: Image Enhancement in Spatial Domain

### 5-Mark Questions

1. Describe contrast stretching and intensity level slicing with examples.
2. Explain histogram equalization.

3. Differentiate between linear and nonlinear spatial filters.
4. Explain median, max, and min filters.
5. What is image magnification? Explain magnification by replication.

### **10-Mark Questions**

1. Explain basic gray-level transformations in detail: negative, log, power-law, and bit-plane slicing.
2. Discuss spatial filtering operations. Explain smoothing and sharpening filters with masks.
3. Compare Sobel, Prewitt, and Roberts operators.
4. Explain high-boost filtering and Laplacian filtering with examples.

## **Unit 3: Frequency Domain Enhancement**

### **5-Mark Questions**

1. Explain the 2D Discrete Fourier Transform.
2. Write short notes on separability and periodicity properties of Fourier Transform.
3. Describe Ideal Low Pass and Butterworth Low Pass Filters.
4. What is a Gaussian High Pass Filter? Explain.
5. Explain convolution, correlation, and padding.

### **10-Mark Questions**

1. Describe the Fourier transform and its properties with mathematical expressions.
2. Compare Ideal, Butterworth, and Gaussian LPF and HPF in the frequency domain.
3. Explain the FFT algorithm for 1-D case and discuss its time complexity.
4. Explain Hadamard, Haar, and DCT transforms with applications.

## **Unit 4: Image Restoration and Compression**

### **5-Mark Questions**

1. Explain Gaussian and impulse noise models.
2. Describe arithmetic, geometric, harmonic, and contraharmonic mean filters.
3. Write short notes on band-pass and band-reject filters.
4. Explain coding redundancy, interpixel redundancy, and psychovisual redundancy.
5. Differentiate between lossless and lossy predictive models.

## **10-Mark Questions**

1. Explain image degradation and restoration model with a block diagram. Describe various noise models.
2. Discuss order statistic filters in detail with examples.
3. Explain Huffman coding, Run Length Coding, and IGS coding scheme.
4. Describe lossless and lossy predictive compression models with block diagrams.

## **Unit 5: Morphological Image Processing**

### **5-Mark Questions**

1. Explain logical operations on binary images.
2. Define ‘fit’ and ‘hit’ in morphology.
3. Explain dilation operation with an example.
4. Explain erosion operation with an example.
5. Differentiate between opening and closing.

### **10-Mark Questions**

1. Describe morphological operations—dilation, erosion, opening, and closing—with diagrams and examples.
2. Explain the role of morphological processing in image enhancement and segmentation applications.

## **Unit 6: Image Segmentation**

### **5-Mark Questions**

1. Define image segmentation and explain similarity vs. discontinuity approaches.
2. Explain point, line, and edge detection.
3. Explain Laplacian-based edge detection and the Mexican hat filter.
4. Describe global and local thresholding.
5. Write short notes on region growing.

## **10-Mark Questions**

1. Discuss gradient-based edge detection operators (Sobel, Prewitt, Roberts) and compare them.
2. Explain Hough Transform for line detection with mathematical formulation.
3. Describe thresholding techniques—global, local, and adaptive—with examples.
4. Explain region growing, region splitting, and region merging algorithms.

## **Unit 7: Representation, Description, and Recognition**

### **5-Mark Questions**

1. What are chain codes? Explain with an example.
2. Write short notes on Fourier descriptors.
3. Define patterns and pattern classes.
4. Explain a simple perceptron model.
5. Explain shape numbers and signatures.

### **10-Mark Questions**

1. Describe chain codes, signatures, Fourier descriptors, and shape numbers in detail.
2. Explain decision-theoretic methods for pattern recognition.
3. Discuss neural-network-based image recognition using a multilayer perceptron.
4. Explain the block diagram and working of a general pattern recognition system.